CIRCUIT DESIGN

PROBLEM STATEMENT ID	PROBLEM STATEMENT	PROBLEM STATEMENT DESCRIPTION
CKTD-01	**Ultra-Low-Power IoT Edge Sensor Node for Rural Environmental Monitoring**	Develop a low-power IoT environmental monitoring system using ESP32/STM32 to measure PM2.5, temperature, and humidity, achieving 30+ days of battery life through deep-sleep mode. The device performs edge-based threshold alerts, transmits data via LoRa/GSM (MQTT) without cloud dependency, and includes a low-cost (<₹2,000) BOM, circuit prototype/simulation, and power consumption analysis.
CKTD-02	**Smart Battery Protection & Active Cell Balancing Module for EV Applications**	Develop an Intelligent Battery Management System (BMS) for a 4-cell lithium-ion battery pack with cell balancing, SoC estimation, and protection against overvoltage, undervoltage, overcurrent, and over-temperature using NTC thermal monitoring and relay cutoff. The system provides CAN/UART data output, includes a working hardware prototype, SoC display, and validated charge/discharge test results with circuit schematics.

CKTD-03	**FPGA-Based Real-Time Signal Processing Accelerator**	Develop an FPGA-based real-time Signal Processing Accelerator featuring a configurable FIR filter bank, FFT module, and lightweight neural network (e.g., LeNet-5) for classification. The project includes VHDL/Verilog implementation, simulation and resource utilization reports (LUTs, DSPs, BRAMs), accuracy evaluation (e.g., MNIST), architecture diagram, and demonstrates lower latency and higher throughput than an equivalent MCU-based software implementation.
CKTD-04	**AI-Assisted Configurable Analog Front-End for Multi-Modal Biomedical Sensing**	Develop a configurable low-noise Analog Front-End (AFE) for ECG, SpO_2/PPG, and temperature sensors with a programmable gain amplifier, configurable filters, MCU-based auto-calibration (offset/gain), and a 12-bit+ ADC interface. The system includes a hardware prototype, firmware for mode switching, a PC/web app displaying live biosignal waveforms, and complete circuit schematics.
CKTD-05	**Hardware-Secured Tamper-Resistant Data Logger for Critical Infrastructure**	Develop a tamper-resistant secure data logger with AES-256 encrypted storage, secure boot, physical tamper detection, RTC-based timestamped logging, and HMAC-authenticated UART/USB data export. The project includes a hardware prototype, encrypted SD card storage, tamper detection circuit, secure

		boot demonstration, and a PC log viewer with integrity verification.
CKTD-06	**Wireless Inductive Power Transfer System with AI-Based Foreign Object Detection**	Develop a wireless inductive power transfer system (100–200 kHz) delivering 5W at 85%+ efficiency over a 5 mm air gap with Foreign Object Detection (FOD) using Q-factor/resonance shift and an ML classifier to distinguish metallic objects from valid receivers. The system includes overvoltage/overcurrent protection, automatic shutdown on FOD detection, and a hardware prototype with transmitter/receiver coils, efficiency testing, and circuit schematics.
CKTD-07	**Reconfigurable Software-Defined Instrumentation Platform on FPGA**	Develop an FPGA-based Reconfigurable Software-Defined Instrumentation Platform that functions as at least three instruments—such as a 2-channel digital oscilloscope (≥1 MSPS), 8-channel logic analyzer (≥10 MHz), arbitrary waveform generator (up to 100 kHz), frequency counter, or SPI/I²C/UART protocol analyzer—with runtime mode switching without reprogramming the FPGA. The project includes HDL implementation, a Python/Qt host PC application, performance evaluation, resource utilization report, and architecture documentation.